CC-TECHNO 32

Stage 1 Validation Report

EnerGreen: Green Energy Consumption Monitoring System

Team Name: GreenTechs

Team Member	**Role (Hustler, Hacker, Hipster)**
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Summary

The initial validation phase for Energreen focused on understanding the pain points of homeowners regarding electricity consumption and identifying their interest in energy-monitoring solutions. With 121 respondents, findings revealed that 76.8% experience expensive bills despite limited appliance usage, with 70.3% struggling to pay electricity bills occasionally or very often. Most respondents employ basic conservation methods like unplugging devices (85.1%) and limiting appliance usage (87.6%), yet only 9.9% track their electricity usage intensively. Notably, 99.1% expressed interest in IoT-based solutions for real-time electricity tracking, with 90.1% willing to invest if affordable (86.7% preferring solutions under ₱3,000). Additionally, 95% showed interest in solar solutions. These findings confirm a clear need and market interest in accessible, real-time energy tracking tools that integrate with solar solutions, validating Energreen's core hypothesis and providing direction for future product development.

Validation Objectives

- **Determine whether homeowners struggle to estimate their electricity usage and how this impacts their financial planning.**
  We aimed to assess whether homeowners have difficulty estimating their electricity consumption and how this impacts their financial planning. This objective was crucial to validate our assumption that consumers lack visibility into their energy usage patterns.

- **Investigate whether homeowner's have access to tools that help them monitor their electricity usage.**
  We sought to understand if homeowners currently have adequate tools to monitor electricity usage. This objective helps determine the gap in the market for effective monitoring solutions.

- **Determine whether IoT integration for real-time electricity tracking is viable, accessible and desirable for users.**
  We wanted to assess whether IoT integration for real-time electricity tracking would be viewed as valuable, accessible, and desirable by potential users. This objective directly informs product development directions for Energreen.

Methodology

The validation was conducted through an **online survey using Google Forms**. A total of **121 respondents** from various income levels (covering low to upper-income homeowners) participated in the survey. The questionnaire included 11 questions focusing on electricity consumption habits, bill management challenges, interest in energy efficiency solutions, and willingness to adopt IoT and solar technologies. Both quantitative (multiple-choice) and qualitative (select-all-that-apply) question formats were utilized to gather comprehensive insights.

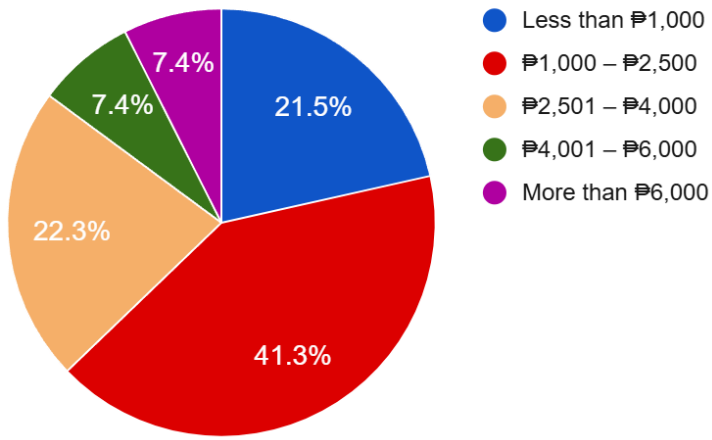
Stage 1 - Validation Board

Startup Name:	Energreen
Customer Segment	Low to Upper Income Homeowners
Hypothesis	These customers struggle to track their electricity consumption, leading to unexpected expenses and inefficient energy use, and they often lack access to affordable solutions for monitoring and optimizing their energy consumption.
Experiment (Method)	Online Survey
Success Metric	At least 70% of survey respondents strongly express difficulty in tracking their electricity consumption, resulting in higher expenses, validating the hypothesis.
Result	The survey shows that 76.8% of respondents report experiencing expensive electricity bills despite limited appliance usage. When it comes to monitoring efforts, only 9.9% track their electricity usage intensively, while the vast majority—87.6%—make only moderate to minimal efforts to manage their consumption. This confirms that even with basic monitoring practices, users still struggle to effectively track and control their electricity usage, which contributes to unexpectedly high expenses.
Action	**PERSEVERE:** Our validation results strongly support our initial hypothesis. We will proceed to Stage 2 validation with a prototype of an affordable IoT energy monitoring system priced under ₱3,000 that offers real-time consumption tracking and potential solar savings visualization.

All Insights

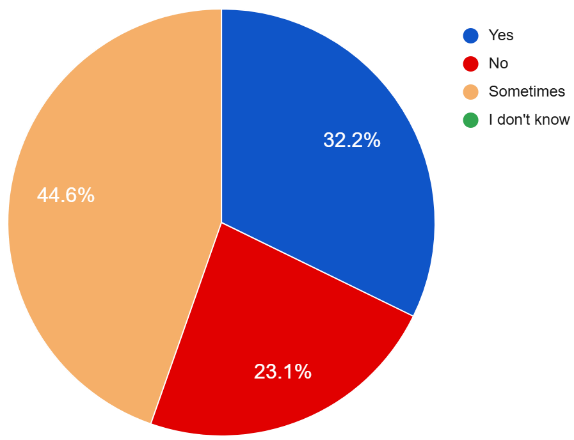
Based on the survey data from 121 respondents, we've identified several key insights:

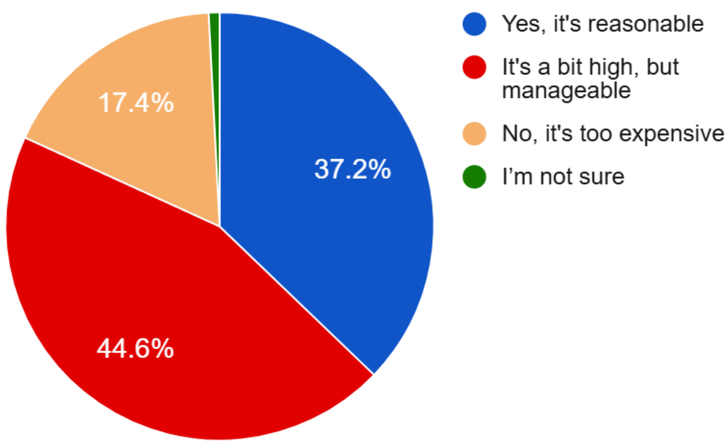
Electricity Expense Distribution:



- 41.3% of respondents pay between ₱1,000-₱2,500 monthly
- 37.1% pay more than ₱2,500 monthly
- Only 21.5% pay less than ₱1,000

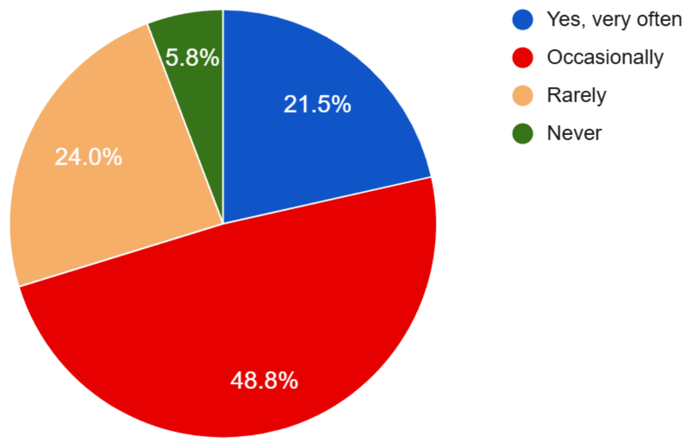
Bill Satisfaction and Perception:





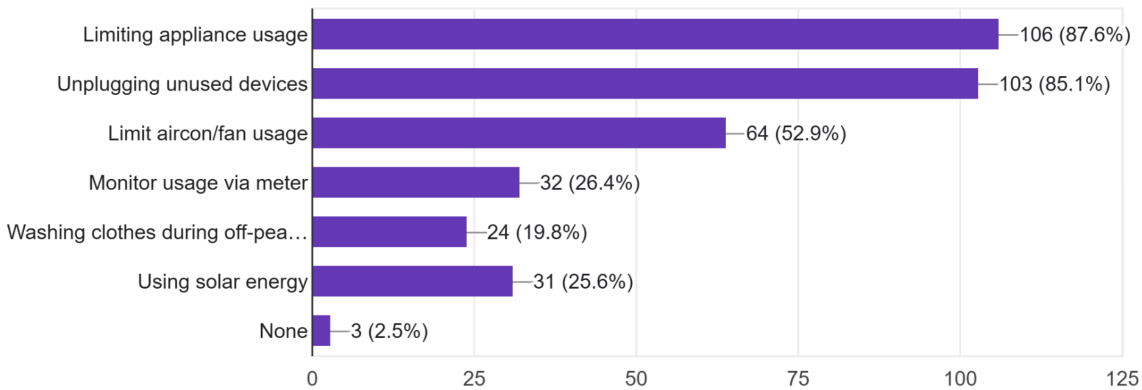
- 76.8% report experiencing expensive bills despite limited appliance usage (32.2% "Yes", 44.6% "Sometimes")
- 62% find their bills either "too expensive" (17.4%) or "a bit high but manageable" (44.6%)
- Only 37.2% consider their current bills reasonable

Payment Difficulties:



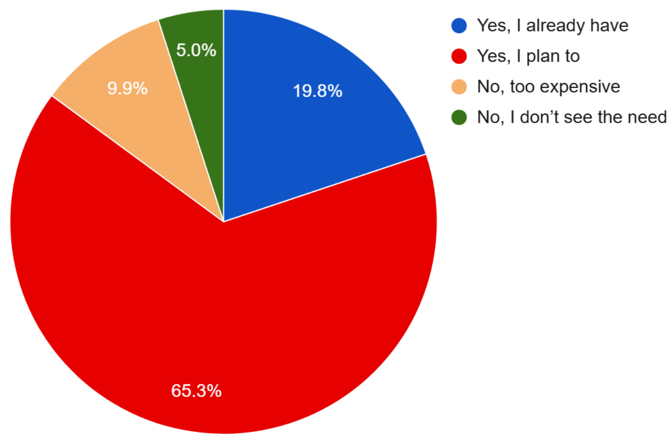
- 70.3% struggle to pay electricity bills either "occasionally" (48.8%) or "very often" (21.5%)
- Only 5.8% never experience payment difficulties

Energy Conservation Practices:



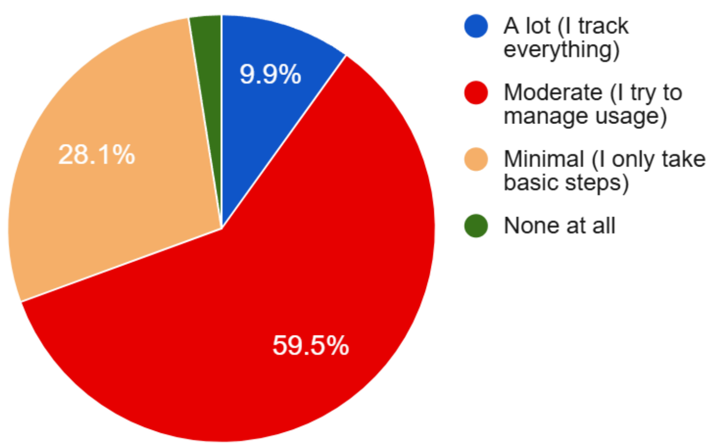
- 87.6% limit appliance usage to reduce costs
- 85.1% unplug unused devices
- 52.9% limit air conditioning/fan usage
- Only 26.4% monitor usage via their meter
- Only 19.8% shift activities to off-peak hours
- Only 25.6% use solar energy

Energy Efficiency Awareness:



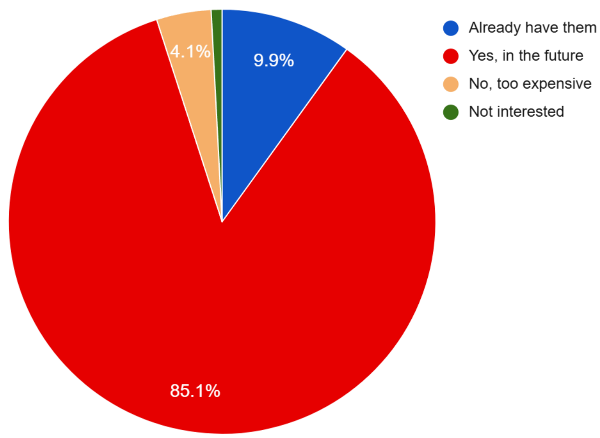
- 85.1% have already adopted (19.8%) or plan to adopt (65.3%) energy-efficient appliances
- Only 9.9% find energy-efficient alternatives too expensive

Monitoring Effort:



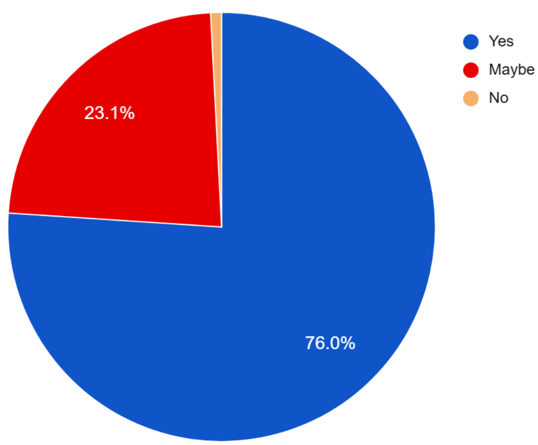
- Only 9.9% track their electricity usage intensively
- 59.5% make moderate efforts to manage usage
- 28.1% take minimal steps
- 2.5% make no effort at all

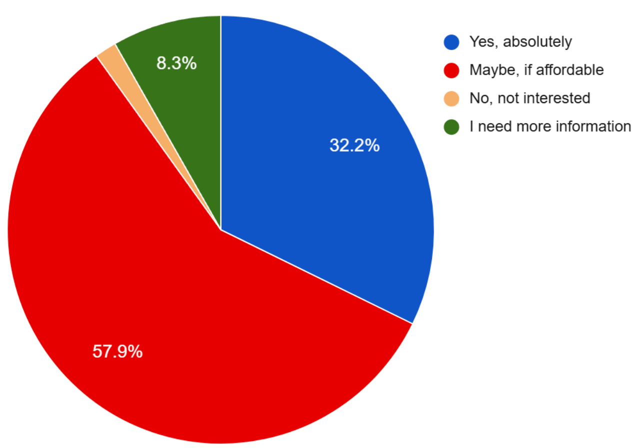
Solar Energy Interest:



- 95% show interest in solar solutions (9.9% already have them, 85.1% plan to invest)
- Only 4.1% find solar solutions too expensive or uninteresting

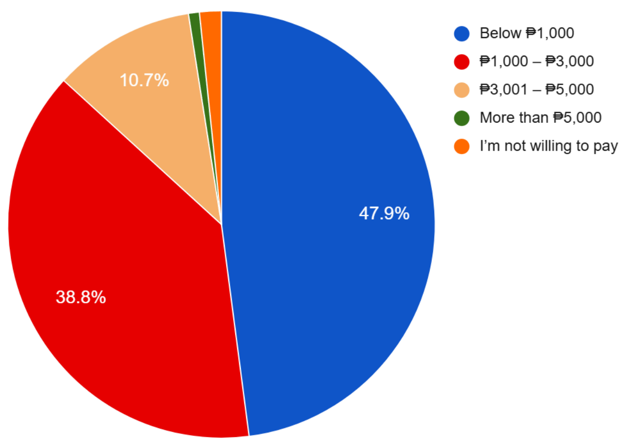
IoT Solutions Interest:





- 99.1% express interest in IoT solutions for real-time electricity tracking (76% "Yes", 23.1% "Maybe")
- Less than 1% have no interest
- 90.1% are open or willing to invest to IoT solutions for monitoring both electricity and solar energy (32.2% "absolutely," 57.9% if affordable)

Price Sensitivity for IoT Solutions:



- 47.9% willing to pay below ₱1,000
- 38.8% willing to pay ₱1,000-₱3,000
- 10.7% willing to pay ₱3,001-₱5,000
- Only 2% unwilling to pay anything

Opportunities

Based on our validation results, we've identified several promising opportunities that align with both user demand and market gaps:

1. **Affordable IoT Monitoring Devices**

   There is a strong market demand for a cost-effective IoT-based electricity monitoring system, with 99.1% of respondents expressing interest and 90.1% open to investing if the price is right. This opens the door to offering a real-time tracking solution under ₱3,000, helping users identify energy waste and reduce costs.

2. **Solar Energy Integration Platform**

With 95% of respondents interested in solar solutions, we see a compelling opportunity to build a hybrid platform that tracks both grid electricity and solar energy production. This would help users visualize savings, optimize energy use, and increase the value of solar investments.

3. **Tiered Product Offerings**

   Given the clear price sensitivity (47.9% willing to pay under ₱1,000 and 38.8% up to ₱3,000), we can offer multiple product tiers:

   - **Basic Tier**: Affordable real-time monitoring
   - **Advanced Tier**: Includes solar tracking, smart tips, and automation. This segmentation allows us to serve a wider market with diverse financial capacities.

4. **Energy Management Mobile App**

   Despite concerns over high electricity bills, only a small portion of users actively monitor their usage. There is an opportunity to introduce a user-friendly mobile app that includes:

   - Real-time consumption data
   - Peak usage alerts
   - Personalized cost-saving tips
   - Monthly reports and budget forecasts

5. **Consumer Education and Engagement Tools**

   With only 10% of respondents tracking their energy use thoroughly, incorporating an educational component into our solution—such as tutorials, gamified challenges, or daily tips—can help increase awareness and drive more responsible energy usage.

Recommendations

To improve our startup idea based on the insights gathered, our plan is as follows:

1. **Develop a Prototype or Wireframe**

   We will create a clickable prototype or wireframe of the Energreen web-based IoT system that showcases core features such as:

   - Basic electricity tracking interface
   - Solar savings projection tools

- Cost breakdowns and peak-hour alerts

This will allow us to test desirability and usability with target users in the next validation stage, ensuring the features we prioritize are those that truly matter to them.

2. **Focus on User Education and Simplicity**

   Since many users make only minimal effort to monitor their energy usage, we will include features in the prototype like:

   - Interactive walkthroughs
   - Educational content
   - Visual insights (e.g., graphs, trends, and usage tips)

   These will serve to validate whether such features can improve awareness and perceived control over electricity consumption.

3. **Explore Tiered Pricing Models**

   Given the price sensitivity observed, we will explore a multi-tiered subscription concept during validation:

   - A Basic Tier with essential monitoring and tips
   - A Premium Tier with solar tracking and advanced analytics

   Rather than committing to a fixed price or full feature set, we'll gather feedback on which features users value enough to pay for and what price ranges feel acceptable.

4. **Conduct Targeted Validation Activities**

   We plan to engage users in low-upper income households or energy-conscious communities to:

   - Test wireframes and gather feature feedback
   - Understand real-world pain points
   - Measure potential impact on user perception and behavior

   This stage will guide our decisions on feature prioritization, pricing structure, and overall product direction before moving into development.

Relevant Links

https://drive.google.com/drive/folders/1h4120c_HFJ7lh3VCHHwc-TzCc5Rtuqlt?usp=sharing